

Nuclear fission and fusion

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

	QUESTION	ANSWER	✓
1	Define nuclear fission.	The splitting of a large, unstable nucleus into two smaller nuclei.	☐
2	Give two examples of elements that can undergo nuclear fission.	Uranium and plutonium	☐
3	Define spontaneous fission.	Fission that happens without the nucleus absorbing a neutron.	☐
4	State what usually has to happen before a nucleus undergoes fission.	The unstable nucleus must absorb a neutron.	☐
5	State what a nucleus splits into during fission.	Two smaller nuclei, two or three neutrons and gamma rays	☐
6	Define a chain reaction.	Neutrons released by one fission cause further fissions, which release more neutrons, and so on.	☐
7	Define nuclear fusion.	The joining of two light nuclei to form a heavier nucleus.	☐
8	In nuclear fusion, which has more mass: the two light nuclei or the nucleus they form?	The two light nuclei	☐
9	Some mass is lost during nuclear fusion. State what it is converted into.	Energy (of radiation)	☐
10	State the type of energy store that all fission products have.	Kinetic energy	☐
11	State how the chain reaction in a reactor differs from that in a nuclear weapon.	Reactor: controlled. Weapon: uncontrolled, causing an explosion.	☐
12	Name the process that releases energy in stars.	Nuclear fusion	☐